

CUSTOMER CHURN BUSINESS ANALYSIS

Technical Report

Project Introduction

Customer churn refers to customers who stop using a company's products or services. High customer churn can negatively affect business growth and revenue.

This project performs an end-to-end analysis of customer churn data to identify important patterns and factors associated with customers leaving the company.

The project includes data validation, data cleaning, exploratory data analysis, statistical analysis, hypothesis testing, confidence interval analysis, and business recommendations.

The purpose of this analysis is to transform raw customer data into meaningful insights that can support better business decision-making and customer retention strategies.

Business Problem

Customer retention is an important challenge for many businesses. When customers leave a company, the business may lose revenue and incur additional costs to acquire new customers.

The main business problem addressed in this project is to understand which customer characteristics are associated with customer churn.

The project attempts to answer the following question:

What factors are associated with customer churn, and how can the business use these insights to improve customer retention?

Project Objectives

The main objectives of this project are:

- Validate and prepare the customer dataset for analysis.
 - Identify patterns associated with customer churn.
 - Perform exploratory data analysis using professional visualizations.
 - Compare customer characteristics between churned and non-churned customers.
 - Perform statistical hypothesis testing.
 - Calculate confidence intervals for important business metrics.
 - Translate analytical findings into actionable business recommendations.
 - Develop a practical customer retention implementation plan.
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Dataset Description

The project uses a customer churn dataset containing **500 customer records and 9 variables**.

The dataset contains the following features:

- **CustomerID** – Unique customer identification number.
- **Tenure** – Duration of the customer's relationship with the company.
- **MonthlyCharges** – Monthly amount charged to the customer.
- **TotalCharges** – Total amount charged to the customer.
- **Contract** – Customer contract type.
- **PaymentMethod** – Method used by the customer for payment.
- **PaperlessBilling** – Whether the customer uses paperless billing.
- **SeniorCitizen** – Indicates whether the customer is a senior citizen.
- **Churn** – Indicates whether the customer left the company.

The dataset was used to analyze relationships between customer characteristics and churn behavior.

Data Cleaning and Preparation

Before performing the analysis, the customer churn dataset was examined to ensure that the data was suitable for analysis.

The following data quality checks were performed:

- Dataset shape was examined.
- Column names and data types were checked.
- Missing values were identified.
- Duplicate records were checked.
- Numerical and categorical variables were reviewed.

Data Quality Results

The dataset contained **500 rows and 9 columns**.

The analysis found:

- **No missing values**
- **No duplicate rows**
- Numerical columns were correctly identified.
- Categorical variables were reviewed for consistency.

Therefore, the dataset was considered suitable for further analysis.

The cleaned dataset was saved as:

cleaned_data.csv

This cleaned dataset was then used for Exploratory Data Analysis and Advanced Statistical Analysis.

Data Preparation Process

The dataset preparation process ensured that the data was organized and ready for analysis.

The main variables used during the analysis included:

- Tenure
- MonthlyCharges
- TotalCharges
- Contract
- PaymentMethod
- Churn

The cleaned dataset was used consistently across all analysis notebooks to maintain reproducibility and ensure reliable results.

```
===== DATASET INFORMATION =====  
  
Shape:  
(500, 9)  
  
Columns:  
Index(['CustomerID', 'Tenure', 'MonthlyCharges', 'TotalCharges', 'Contract',  
       'PaymentMethod', 'PaperlessBilling', 'SeniorCitizen', 'Churn'],  
      dtype='object')  
  
Data Types:  
CustomerID      object  
Tenure          int64  
MonthlyCharges  int64  
TotalCharges    int64  
Contract        object  
PaymentMethod   object  
PaperlessBilling object  
SeniorCitizen   int64  
Churn           int64  
dtype: object  
  
Missing Values:  
CustomerID      0  
Tenure          0  
MonthlyCharges  0  
TotalCharges    0  
Contract        0  
PaymentMethod   0  
PaperlessBilling 0  
SeniorCitizen   0  
Churn           0  
dtype: int64  
  
Duplicate Rows:  
0
```

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand customer behavior, identify churn patterns, and examine relationships between important variables.

The analysis included customer churn distribution, contract type analysis, payment method analysis, numerical data distributions, scatter plot analysis, and correlation analysis.

A total of **seven visualizations** were created.

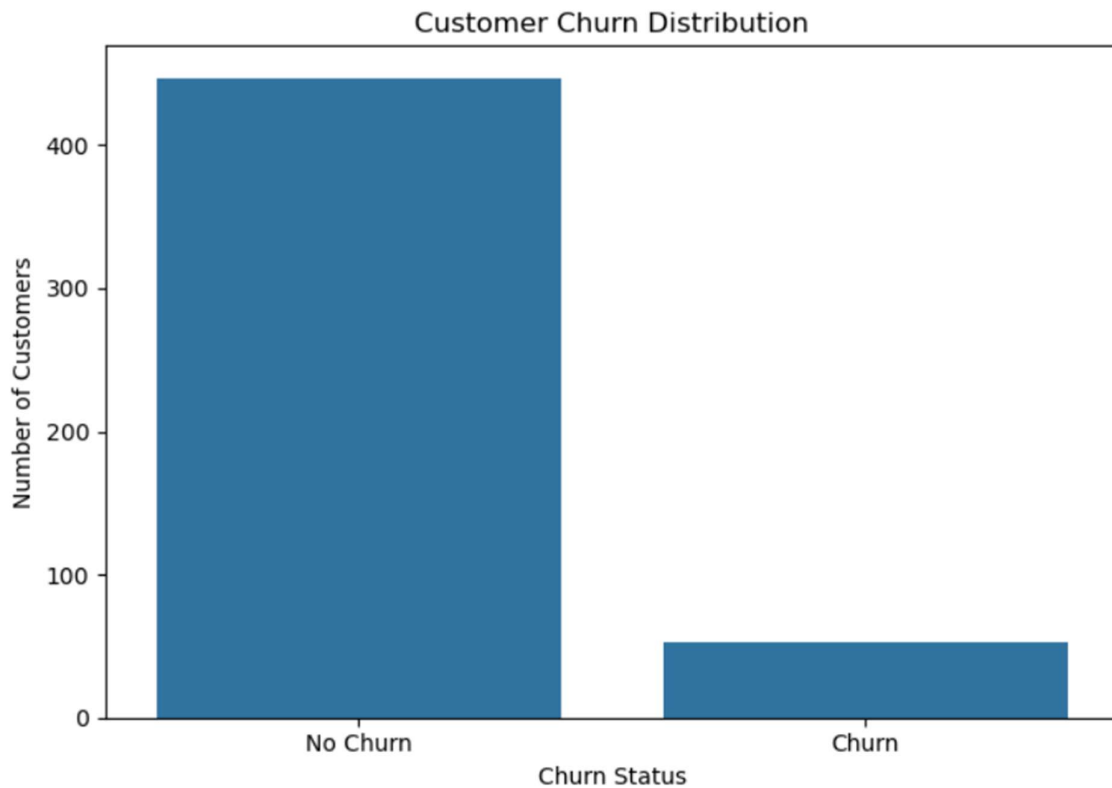
7.1 Customer Churn Distribution

The customer churn distribution was analyzed to understand the overall proportion of customers who left the company.

Out of 500 customers:

- **447 customers did not churn**
- **53 customers churned**
- Overall churn rate was **10.6%**

This shows that although most customers remained with the company, customer churn still represents an important business concern.



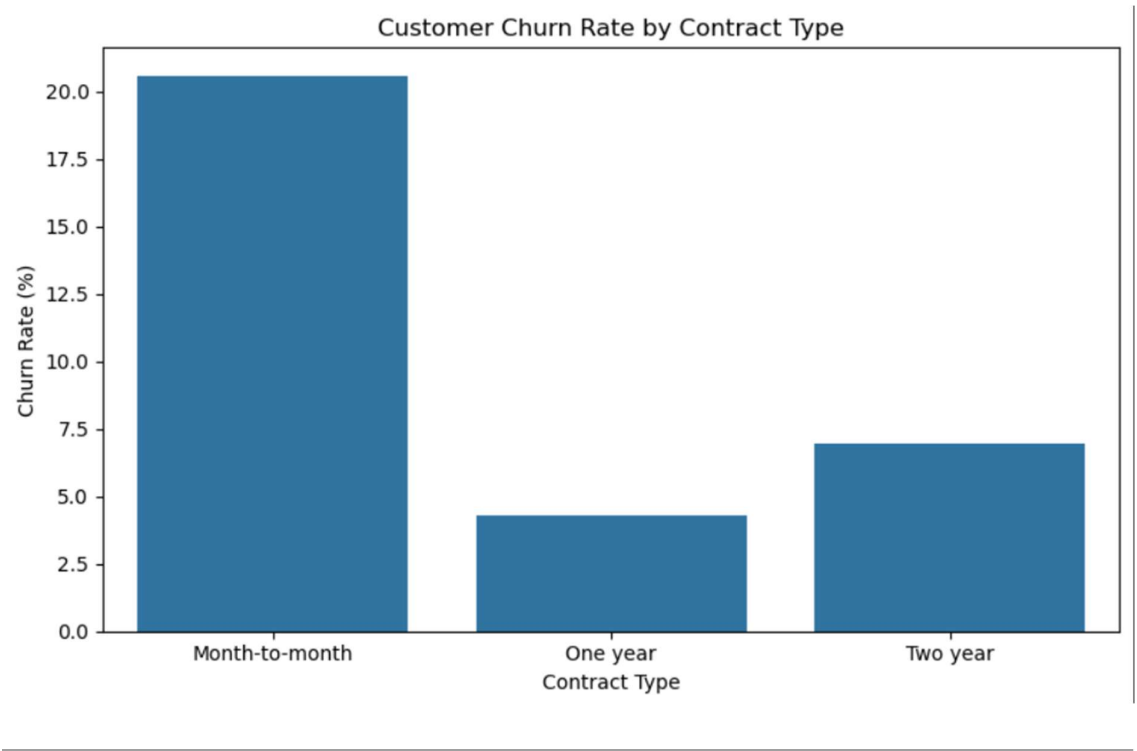
7.2 Churn Rate by Contract Type

Customer churn was compared across different contract types.

- **Month-to-month:** 20.59%
- **One-year:** 4.30%
- **Two-year:** 6.94%

Month-to-month customers had the highest observed churn rate.

This suggests that customers without longer-term contracts may have a higher risk of leaving the company.



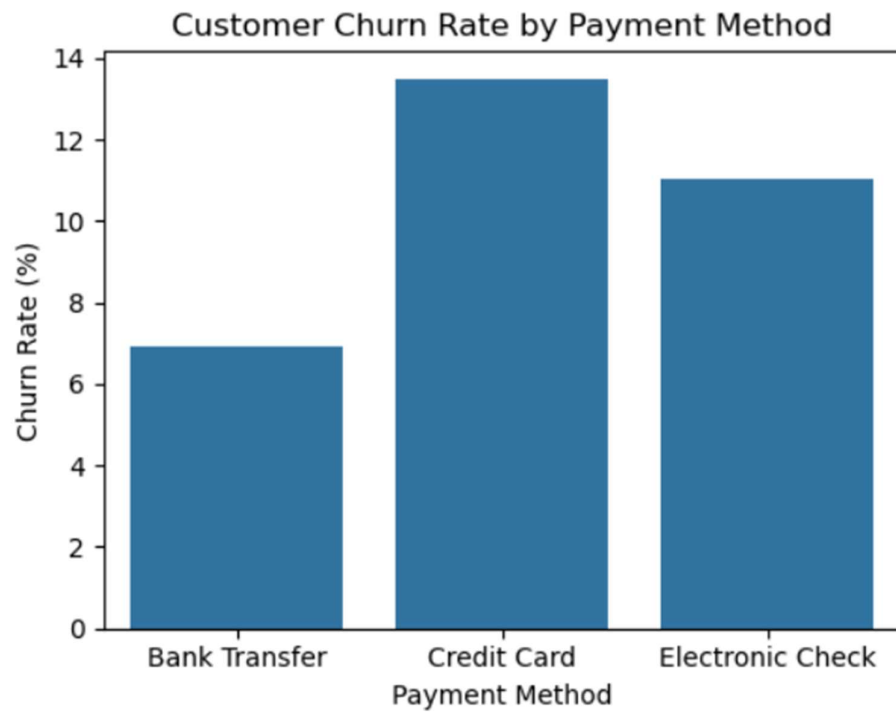
7.3 Churn Rate by Payment Method

The churn rate was also analyzed according to customer payment methods.

- **Bank Transfer:** 6.92%
- **Credit Card:** 13.48%
- **Electronic Check:** 11.04%

Customers using Credit Card had the highest observed churn rate, while Bank Transfer customers had the lowest.

This suggests that payment method may be associated with differences in customer churn behavior.

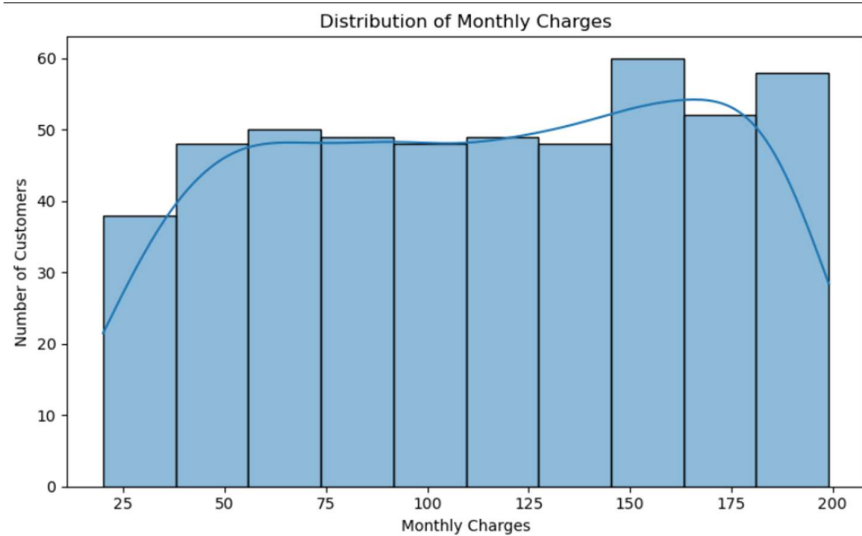


7.4 Monthly Charges Distribution

The distribution of Monthly Charges was analyzed using a histogram.

This visualization helped identify how customer charges were distributed across different price ranges.

Further analysis compared Monthly Charges between churned and non-churned customers.

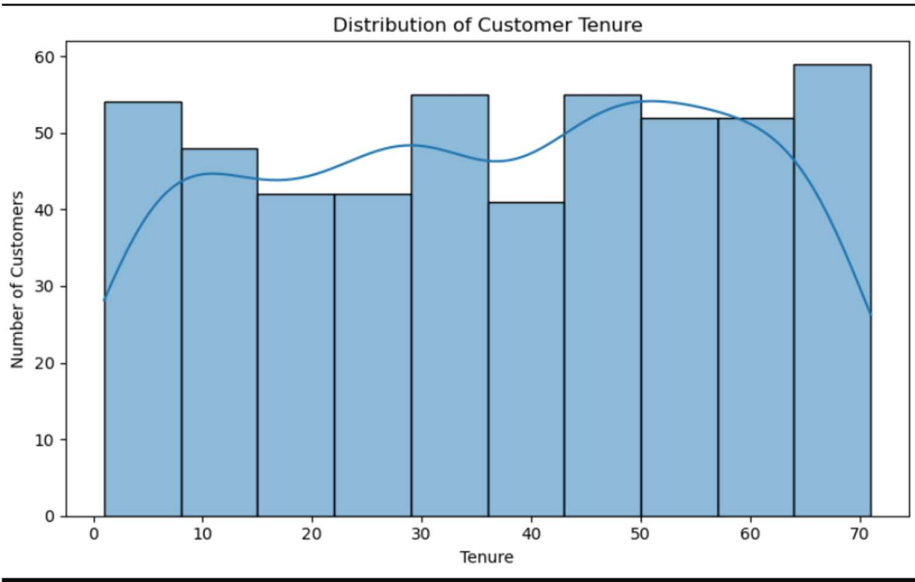


7.5 Customer Tenure Distribution

Customer tenure was analyzed to understand how long customers had been associated with the company.

The tenure distribution provided insight into the duration of customer relationships.

Further statistical analysis showed that customer tenure had an important association with churn.

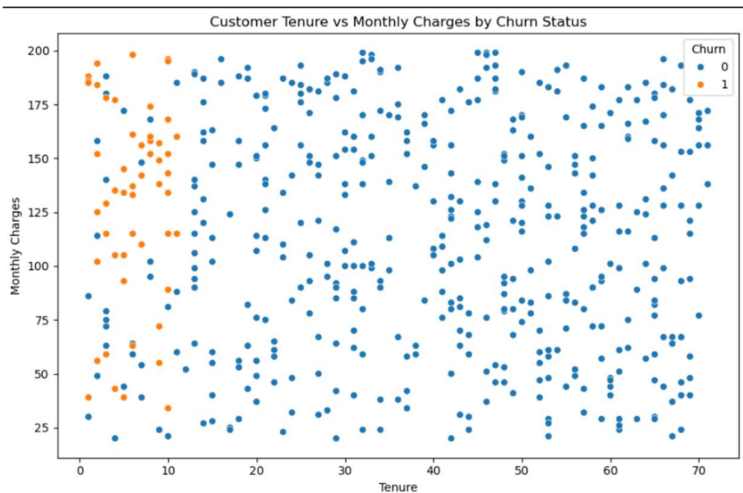


7.6 Tenure vs Monthly Charges

A scatter plot was created to examine the relationship between customer tenure and Monthly Charges based on churn status.

This visualization helped compare patterns between churned and non-churned customers.

The results suggested that churn behavior may differ depending on customer tenure and Monthly Charges.



7.7 Correlation Analysis

A correlation heatmap was created to examine linear relationships between numerical variables.

The numerical variables included:

- Tenure
- MonthlyCharges
- TotalCharges
- SeniorCitizen
- Churn

Correlation analysis helps identify associations between variables but does not prove causation.



Statistical Analysis and Hypothesis Testing

Statistical Analysis

Statistical analysis was performed to determine whether the differences observed between churned and non-churned customers were statistically significant.

The analysis included:

- Descriptive statistics by churn group
- Independent sample t-tests

- Confidence interval analysis

A significance level of **0.05** was used for hypothesis testing.

If the p-value was less than 0.05, the result was considered statistically significant.

8.1 Descriptive Statistics by Churn Status

The numerical characteristics of churned and non-churned customers were compared.

Customer Tenure

- Average tenure of non-churned customers: **40.15**
- Average tenure of churned customers: **6.00**

This shows a major difference between the two groups.

Monthly Charges

- Average Monthly Charges of non-churned customers: **111.72**
- Average Monthly Charges of churned customers: **129.77**

Churned customers had higher average Monthly Charges.

Total Charges

- Average Total Charges of non-churned customers: **4234.58**
- Average Total Charges of churned customers: **4265.75**

The average Total Charges were relatively similar between the two groups.

===== DESCRIPTIVE STATISTICS BY CHURN =====									
Churn	Tenure					MonthlyCharges			
	mean	median	std	min	max	mean	median	std	\
0	40.152125	42.0	18.783141	1	71	111.722595	113.0	52.130132	
1	6.000000	6.0	3.162278	1	11	129.773585	137.0	46.298630	
Churn	TotalCharges								
	min	max	mean	median	std	min	max		
0	20	199	4234.577181	4096.0	2261.912833	159	7992		
1	34	198	4265.754717	4534.0	2271.043533	458	7858		

8.2 Hypothesis Test 1: Tenure vs Customer Churn

Null Hypothesis (H₀)

There is no significant difference in average customer tenure between churned and non-churned customers.

Alternative Hypothesis (H_1)

There is a significant difference in average customer tenure between churned and non-churned customers.

Results

- **T-statistic:** -34.53
- **P-value:** 8.98×10^{-130}
- **Result:** Statistically Significant

Since the p-value is much smaller than 0.05, the null hypothesis was rejected.

There is a statistically significant difference in customer tenure between churned and non-churned customers.

This indicates that **customer tenure is strongly associated with churn in this dataset.**

```
===== HYPOTHESIS TEST 1 =====  
Test: Tenure Difference by Churn Status  
T-statistic: -34.53490213270181  
P-value: 8.982326405691704e-130  
Result: Statistically Significant  
Reject the Null Hypothesis.
```

8.3 Hypothesis Test 2: Monthly Charges vs Customer Churn

Null Hypothesis (H_0)

There is no significant difference in average Monthly Charges between churned and non-churned customers.

Alternative Hypothesis (H_1)

There is a significant difference in average Monthly Charges between churned and non-churned customers.

Results

- **T-statistic:** 2.65
- **P-value:** 0.0101
- **Result:** Statistically Significant

Since the p-value is less than 0.05, the null hypothesis was rejected.

There is a statistically significant difference in Monthly Charges between churned and non-churned customers.

This suggests that **Monthly Charges are associated with customer churn** in this dataset.

```
===== HYPOTHESIS TEST 2 =====  
Test: Monthly Charges Difference by Churn Status  
T-statistic: 2.646439327283195  
P-value: 0.010079197699533079  
Result: Statistically Significant  
Reject the Null Hypothesis.
```

8.4 Hypothesis Test 3: Total Charges vs Customer Churn

Null Hypothesis (H_0)

There is no significant difference in average Total Charges between churned and non-churned customers.

Alternative Hypothesis (H_1)

There is a significant difference in average Total Charges between churned and non-churned customers.

Results

- **T-statistic:** 0.09
- **P-value:** 0.9250
- **Result:** Not Statistically Significant

Since the p-value is greater than 0.05, we fail to reject the null hypothesis.

There is no statistically significant difference in Total Charges between churned and non-churned customers.

```
===== HYPOTHESIS TEST 3 =====  
Test: Total Charges Difference by Churn Status  
T-statistic: 0.09453830507131283  
P-value: 0.9249730721139848  
Result: Not Statistically Significant  
Fail to Reject the Null Hypothesis.
```

8.5 Confidence Interval Analysis

A 95% confidence interval was calculated for the average Monthly Charges.

Results

- **Mean Monthly Charges:** 113.64
- **Margin of Error:** ± 4.55
- **95% Confidence Interval:** 109.08 to 118.19

Based on the sample data, the underlying population mean Monthly Charges is estimated to fall within this range with 95% confidence.

```
===== 95% CONFIDENCE INTERVAL =====  
Mean Monthly Charges: 113.636  
Margin of Error: 4.5514176289484425  
Lower Bound: 109.08458237105155  
Upper Bound: 118.18741762894844  
95% Confidence Interval: (109.08, 118.19)
```

Key Findings

Key Findings

The analysis produced several important findings related to customer churn.

1. Customer Churn Rate

Out of 500 customers, 53 customers churned, resulting in an overall churn rate of **10.6%**.

2. Customer Tenure

Customer tenure showed the strongest relationship with churn.

Churned customers had an average tenure of **6.00**, while non-churned customers had an average tenure of **40.15**.

The difference was statistically significant.

3. Contract Type

Month-to-month customers had the highest churn rate at **20.59%**.

One-year contract customers had the lowest observed churn rate at **4.30%**.

4. Monthly Charges

Churned customers had higher average Monthly Charges (**129.77**) compared to non-churned customers (**111.72**).

This difference was statistically significant.

5. Payment Method

Churn rates varied across payment methods.

- Bank Transfer: **6.92%**
- Credit Card: **13.48%**
- Electronic Check: **11.04%**

6. Total Charges

Total Charges did not show a statistically significant difference between churned and non-churned customers.

10. Business Recommendations

Recommendation 1: Focus on Early Customer Retention

The analysis showed that customers with low tenure have a higher association with churn.

The business should develop strong onboarding programs and engagement strategies during the first few months of the customer relationship.

Recommendation 2: Encourage Longer-Term Contracts

Month-to-month customers showed the highest churn rate.

The company can encourage customers to choose longer-term contracts by offering:

- Discounts
 - Loyalty benefits
 - Special offers
 - Contract upgrade incentives
-

Recommendation 3: Review Pricing Strategies

Churned customers had higher average Monthly Charges.

The company should review whether customers receiving higher charges are receiving sufficient value from the service.

Possible strategies include personalized pricing plans and targeted discounts.

Recommendation 4: Improve Payment Experience

Different payment methods showed different churn rates.

The company should investigate customer payment experiences and identify possible areas of improvement.

Recommendation 5: Develop a Churn Monitoring System

The company should regularly monitor important customer characteristics such as:

- Customer tenure
- Monthly Charges
- Contract type
- Payment method
- Churn trends

Customers identified as high-risk could receive proactive retention offers.

11. Business Implementation Plan

Phase 1: Identify High-Risk Customers

Identify customers who show important churn-risk indicators, particularly:

- Low tenure
- Month-to-month contracts
- Higher Monthly Charges

Phase 2: Retention Campaigns

Develop targeted retention campaigns such as:

- Welcome programs
- Personalized offers
- Loyalty rewards
- Contract upgrade incentives

Phase 3: Monitor Performance

The company should regularly track:

- Customer churn rate
- Customer retention rate
- Contract conversion rate
- Campaign performance

Phase 4: Continuous Improvement

The retention strategy should be regularly evaluated using new customer data.

Business decisions should be updated based on customer behavior and future analytical findings.

12. Conclusion

Conclusion

This project conducted an end-to-end business analysis of customer churn using a dataset containing **500 customer records**.

The project included data validation, data cleaning, exploratory data analysis, statistical hypothesis testing, confidence interval analysis, and business recommendations.

The analysis identified **customer tenure, contract type, and Monthly Charges** as important factors associated with customer churn.

The strongest finding was the substantial difference in customer tenure between churned and non-churned customers. Customers who churned had significantly lower average tenure.

The findings suggest that businesses should prioritize early customer engagement, encourage longer-term contracts, review pricing strategies, and monitor customer churn risk proactively.

This project demonstrates how data analysis can transform raw customer data into meaningful business insights and actionable recommendations.
